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Power Rules and Derivative Rules

Power Rules:

$$\Rightarrow \text{if } f(x) = x^n$$

$$\therefore \frac{d}{dx}(x^n) = nx^{n-1}$$

Derivative Rule:

$$\Rightarrow \frac{d}{dx}(c) = 0$$

$$\Rightarrow \frac{d}{dx}(cf(x)) = c \frac{d}{dx}(f(x))$$

$$\Rightarrow (f+g)' = f' + g'$$

$$(f-g)' = f' - g'$$

Eqⁿ of tangent of Polynomials:

$\Rightarrow$ if $y = f(x)$ and if we need tangent at $x = a$
then,

$$y - y_1 = m(x - x_1)$$

where, (x_1, y_1) lies on the curve

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Derivatives of Trigo. Fuⁿ:

Fu ⁿ	Derivative
$\sin x$	$\cos x$
$\cos x$	$-\sin x$
$\tan x$	$\sec^2 x$
$\cot x$	$-\operatorname{cosec}^2 x$
$\sec x$	$\sec x \tan x$
$\operatorname{cosec} x$	$-\operatorname{cosec} x \cot x$

$$\Rightarrow \frac{d}{dx}(\ln x) = \frac{1}{x}$$

$$\Rightarrow \frac{d}{dx}(e^x) = e^x$$

$$\Rightarrow \frac{d}{dx}(a^x) = a^x \ln a$$

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